

Pragalbha Patil

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Summary

Specialist Software Engineer at Nasdaq with 6+ years of experience building large-scale data platforms, distributed systems, and developer-facing infrastructure. Experienced in market data delivery, Databricks-based data onboarding, API platforms, and AI integrations including MCP-enabled workflows for LLMs. Strong background in backend engineering, cloud-native architectures, observability, and secure access control systems using OAuth2, OIDC, JWT, entitlements, rate limiting, and usage metering. Hands-on across Ruby on Rails, Node.js, Java, AWS, GCP, Kubernetes, Spark, and modern data platform technologies.

Skills

Languages: Ruby, Python, Java, PHP, JavaScript, TypeScript, SQL

Frameworks & Libraries: Ruby on Rails, React, Next.js, Redux, Node.js, Express, NestJS, Spring Boot, Django, Flask, Laravel

Cloud & DevOps: AWS (EKS, ECS, EC2, S3, CloudFront, RDS, IAM, CloudWatch), GCP (GKE, Pub/Sub, BigQuery), Docker, Kubernetes, Terraform, Helm, GitHub Actions, Jenkins, NGINX, CDN

Databases: PostgreSQL, MySQL, MongoDB

APIs & Security: REST, GraphQL, WebSockets, Streaming API, OpenAPI/Swagger, OAuth2, OIDC, JWT, SSO, RBAC, Entitlements, Rate Limiting, Usage Metering

Observability & QA: OpenTelemetry, Prometheus, Grafana, ELK Stack, Datadog, Sentry, RSpec, Jest, pytest, Cypress, Playwright

Domain Expertise: Market Data Delivery, Fintech, Multi-Tenant SaaS, SLA/SLO, Data Onboarding

Experience

Specialist Software Engineer, Nasdaq – Bangalore, IN Apr 2026 – Present

- Leading development of internal AI platform capabilities enabling LLM-driven interaction with Nasdaq Data Link datasets and services
- Designed and built MCP-based integrations allowing AI systems to securely access datasets, metadata, analytics workflows, and internal services
- Architected entitlement-aware AI workflows by integrating OAuth2, OIDC, JWT validation, and usage controls into AI-assisted access patterns
- Improved platform observability through OpenTelemetry instrumentation, tracing, metrics, dashboards, and operational tooling
- Contributed to platform architecture decisions across AI initiatives, developer tooling, and data services

Senior Software Engineer, Nasdaq – Bangalore, IN May 2024 – Apr 2026

- Designed and implemented internal MCP server enabling AI models (LLMs) to interact with Nasdaq Data Link (data.nasdaq.com) APIs, Databricks, and internal data services
- Built table creation and dataset onboarding workflows on Databricks using NDL platform, improving research dataset availability and quality
- Developed automated market data ingestion and ETL pipelines, reducing client onboarding time by 30% and ensuring data validation and SLA compliance
- Enhanced Data Link API integrations (REST and WebSocket) for low-latency data delivery, adding entitlements, rate limiting, and usage metering controls
- Integrated Pendo analytics into Data Link web platform to track SDK (Python/Java/R) adoption, API usage patterns, and developer experience metrics
- Maintained and upgraded Ruby on Rails microservices for market data delivery, improving scalability,

observability (OpenTelemetry, Prometheus, Grafana), and performance under high-throughput workloads

Senior Software Engineer, GoComet – Bangalore, IN

June 2023 – May 2024

- Led enhancements to a multi-tenant analytics dashboard built with React/Redux, Node.js, and REST APIs, processing millions of trade records to deliver real-time P&L, market trend visualizations, and client-specific KPIs
- Optimized backend services in Node.js and Ruby on Rails, improving query performance and API response times by 25% through caching (Redis) and database indexing (PostgreSQL)
- Expanded authentication and authorization features using OAuth2, JWT, and role-based access controls, ensuring secure data access across web and API channels
- Delivered CI/CD improvements with GitHub Actions and Docker, reducing deployment time and improving release reliability
- Collaborated with product and data teams to integrate external APIs, standardize data ingestion pipelines, and ensure SLA compliance

Software Engineer, GoComet – Bangalore, IN

July 2021 – June 2023

- Developed and maintained a high-volume analytics dashboard for supply chain and logistics insights, integrating market data APIs and SQL data warehouses
- Built and refactored backend microservices (Ruby on Rails, Node.js) supporting millions of daily logins via web and API interfaces, with load balancing and horizontal scaling on AWS
- Implemented API documentation with OpenAPI/Swagger to streamline developer onboarding and reduce integration errors
- Introduced automated testing suites (RSpec, Jest) and monitoring (Prometheus, Grafana) to improve platform stability and observability
- Worked closely with stakeholders to gather requirements, design features, and iterate in Agile/Scrum sprints

Education

University of Mumbai - K.J. Somaiya, Bachelor of Engineering - BE, Information Technology

August 2018 – July 2021

- **Coursework:** Computer Architecture, Algorithms, Computational Theory, Machine Learning

Vivekanand Education Society's Polytechnic, High School Diploma in Computer Technology

July 2015 – June 2018

- **Coursework:** Computer Architecture, Algorithms and Programming

Publications

Predictive Policing for Women's safety - SSRN

June 2021

Vishwa Gosalia, *Pragalbha Patil*, Khushboo Mehta - [dx.doi.org/10.2139/ssrn.3867200](https://doi.org/10.2139/ssrn.3867200)

Projects

Orbit

<https://orbit.pragal.fun>

- Developed a real-time multiplayer strategy game leveraging Cloudflare Workers, Durable Objects, D1, Queues, and edge-native storage primitives.
- Designed deterministic game execution and server-authoritative state transitions to ensure consistency across concurrent players.
- Built low-latency presence and synchronization systems using Durable Objects while maintaining operational simplicity through a serverless architecture.
- Engineered leaderboard integrity and anti-cheat mechanisms by validating all gameplay actions on trusted backend services.
- Demonstrates practical application of distributed systems concepts including consistency, coordination, fault recovery, and event-driven processing.